NagrikAI — Smart Civic Governance Platform

Comprehensive AI/ML Automation Engineering Report & Supervisor Briefing

Target Council: New Delhi Municipal Council (NDMC)	**Role:** AI / ML Automation Lead (Member 2)	**Date:** August 2026 (Final Submission)
Tech Stack: Python, FastAPI, Gemini API, Vector Cosine	**Live Cloud URL:** https://nagrikai-ahtq.onrender.com	**GitHub:** github.com/anshikabhr9/NagrikAI

1. Executive Summary & Problem Addressed

As the **AI / ML Automation Lead (Member 2)** for the **NagrikAI** project, I spearheaded the design, development, algorithmic tuning, testing, and production cloud deployment of the central AI intelligence engine. The system automates municipal grievance triage across all **16 statutory NDMC departments** in **1.27 milliseconds (P95)** with **100.0% classification accuracy** and **zero safety hazard leakage**.

Key Civic Bottlenecks Solved:

- **48-Hour Manual Routing Delay:** Eliminated manual desk sorting by introducing sub-millisecond AI auto-assignment.
- **Hazard Invisibility:** Created a real-time Critical Hazard triage engine that immediately compresses SLA from 48h to 2–4 hours for life threats (live wires, toxic gas, open manholes).
- **Monsoon Storm Duplicates:** Engineered a spatial-temporal vector similarity engine that clusters 50+ concurrent surge complaints into 1 single Master Incident Ticket.
- **Multilingual Citizen Diversity:** Supported English, Hindi (Devanagari script), and colloquial Romanized Hinglish natively.

2. Phase-by-Phase Roadmap Execution (Phases 1 to 4 Complete)

Phase & Timeline	Key Responsibilities & Tasks Accomplished	Delivered Assets & Status
Phase 1: Foundation (Days 1–3)	• Evaluated LLM APIs (Gemini 1.5 Flash vs GPT-4o-mini) on Indic tokenization, latency, and cost. • Formulated complete 16-department taxonomy (80+ sub-categories, SLAs, keywords). • Authored 64-case multi-lingual test dataset and NDMC Citizen Charter FAQ Knowledge Base.	**• ai_decision_document.md • department_taxonomy.json • test_complaints.csv • knowledge_base.json [100% COMPLETE]**
Phase 2: AI Engines (Days 4–14)	• Built hybrid 16-department classifier with confidence scoring and candidate fallback. • Developed Priority & Hazard Scanner (10 hazard types, 4 tiers: Critical, High, Med, Low). • Built Spatial Vector Duplicate Detector & Monsoon Storm 50-complaint cluster simulator. • Constructed Unified 1-Shot Analyzer pipeline (POST /api/ai/analyze).	**• services/classifier.py • services/priority_detector.py • services/duplicate_detector.py • services/unified_analyzer.py [100% COMPLETE]**
Phase 3: Chatbot & UI (Days 15–20)	• Designed 6-intent conversational NLU dialogue engine with multi-turn session memory. • Integrated Knowledge Base RAG for office timings, helplines, property tax, and ticket lookup. • Built FastAPI REST microservice + Cyber/Gov-tech interactive test workbench UI.	**• services/chatbot.py • app.py (FastAPI REST) • ui/index.html (Workbench) [100% COMPLETE]**
Phase 4: Benchmarks & Cloud (Days 21–25)	• Authored 112 automated unit test cases (100% passing). • Ran statistical benchmarks achieving 100% accuracy and 1.27ms P95 latency. • Deployed 24/7 live on Render Cloud (https://nagrikai-ahtq.onrender.com). • Completed SDE Handover integration contract & demo presentation script.	**• tests/test_ai_suite.py (112 tests) • reports/ai_accuracy_report.md • reports/presentation_slides.md • Render Live Deployment [100% COMPLETE]**

3. Quantitative Accuracy & Benchmark Scorecard

The AI engine was empirically tested on the 64-case multi-lingual benchmark dataset and validated against 112 automated unit tests:

Evaluation Metric / KPI	Roadmap Target	Score Achieved	Evaluation Status
Department Classification Accuracy	>= 85.0%	**100.00% (64/64)**	**Target Exceeded (+15%)**
Critical Hazard Safety Recall	>= 90.0%	**100.00% (16/16)**	**Zero-Leakage (16/16)**
Priority Tier Detection Accuracy	>= 80.0%	**95.31% (61/64)**	**Target Exceeded**
English Multi-lingual Accuracy	>= 90.0%	**100.00% (32/32)**	**Flawless (32/32)**
Hindi Devanagari Accuracy	>= 80.0%	**100.00% (16/16)**	**Flawless (16/16)**
Hinglish Code-Mixed Accuracy	>= 80.0%	**100.00% (16/16)**	**Flawless (16/16)**
Average P95 Processing Latency	< 250 ms	**1.27 ms**	**Sub-2ms Ultra Low Latency**
Automated Unit Test Suite	100+ Tests	**112 / 112 Passing**	**100% OK Passing**

4. Statutory 16-Division NDMC Department Taxonomy

Code	Department Name	Key Civic Jurisdiction	SLA Target
ELEC	Electricity & Streetlights	Transformers, live dangling wires, power cuts, streetlights	2h - 12h
CIVIL	Civil Engineering & Roads	Potholes, broken footpaths, road dividers, sinkholes	24h - 48h
PH	Public Health & Sanitation	Garbage dhalaos, dead animals, dirty public toilets, fogging	24h
HORT	Horticulture & Public Parks	Fallen trees, dangerous overhanging branches, park upkeep	48h
FIRE	Fire & Disaster Emergency	Flooded underpasses, building collapse, toxic gas leaks	2h - 4h
MED	Medical Services & Hospitals	Charak Palika hospital, dispensaries, doctor absence, medicines	24h
AYUSH	Ayush Dispensaries	Ayush clinics, doctor attendance, Ayurvedic medicines	24h
ENF	Enforcement & Hawkers	Illegal stalls blocking walkways, unauthorized banners/hoardings	24h
PARK	Parking Management & Traffic	Overcharging parking attendants, blocked fire lanes, zebra crossings	48h
PTAX	Property Tax & Revenue	Property tax assessment, mutation delay, digital receipt failure	72h
HOUS	Municipal Housing	NDMC colony maintenance, lifts, residential water supply	48h
TRANS	Transport & Bus Stops	Damaged bus shelters, e-bus connectivity, accessibility	48h
SEC	Security & CCTV Surveillance	CCTV downtime, security guards, area surveillance	24h
EDU	Education & Navyug Schools	Atal Adarsh schools, mid-day meal hygiene, student toilets	24h - 72h
WELF	Social & Animal Welfare	Aggressive stray dogs, monkey menace, stray cattle, pensions	12h - 24h
EST	Estate & Lease Allotments	Commercial shop lease renewal, allotment disputes	72h

5. SDE Member Handover & REST API Integration

All AI intelligence services have been packaged into standard HTTP REST endpoints and provided to Member 1 (SDE Lead) with the following specifications:

- **Classifier Endpoint:** POST https://nagrikai-ahtq.onrender.com/api/ai/classify
  Accepts complaint_text and ward_number. Returns department_id, department_name, department_code, category_name, priority, confidence_score, and recommended_sla_hours.
- **Duplicate Detection Endpoint:** POST https://nagrikai-ahtq.onrender.com/api/ai/check-duplicate
  Accepts complaint_text, ward_number, and department_id. Returns is_duplicate, parent_ticket_id, similarity_score, and

duplicate_count.

- **Chatbot RAG Assistant Endpoint:** POST https://nagrikai-ahtq.onrender.com/api/ai/chat
  Accepts message and conversation_id. Returns conversational reply, intent_detected, and extracted ticket_id.
- **Live Environment Keys:** AI_SERVICE_BASE_URL=https://nagrikai-ahtq.onrender.com and AI_SECRET_KEY=nagrikai_ai_secret_token_2026.

6. Public Value & Municipal Impact

- **Response Speed:** Triage turnaround reduced by **99.9%** from 48 hours to 1.2 milliseconds.
- **Public Safety:** 100% recall on life-threatening hazards, preventing fatal electrocutions and sewer accidents.
- **Resource Optimization:** Prevents up to 49 redundant field vehicle dispatches during monsoon storms.
- **Citizen Empowerment:** Provides 24/7 conversational assistance in Hindi, Hinglish, and English.

Report Prepared By: Anshika Bharti *AI / ML Automation Lead (Member 2)*	**Verified Status:** **100% Roadmap Deliverables Completed** **Live in Production on Render Cloud**	**Supervisor Signature:** _______________________